Zeynep Yıldız

MSc Student in Data Science & AI | Computer Science & Finance Graduate

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Education

Bogazici University | Istanbul, Turkiye (Oct 2024 – Exp. Jan 2027)
MSc. in Data Science and Artificial Intelligence
Thesis: Internalizing Investment Mandates: Neurosymbolic Reinforcement Learning for Regime-Aware Asset Allocation
Advisors: Ercan Atam (Bogazici University) & Erman Acar (University of Amsterdam)
GPA: 3.57

Sabanci University | Istanbul, Turkiye (Oct 2020 – Jun 2024)
Faculty of Engineering and Natural Sciences
Major: Computer Science & Engineering | Minor: Finance
50% Scholarship Cumulative GPA: 3.32

Publications

- **Z. Yıldız**, S. Chatterji, E. Atam, E. Acar. *Internalizing Investment Mandates: A Neurosymbolic RL Approach to Portfolio Allocation.* In Proceedings of the 7th ACM International Conference on AI in Finance (ICAIF '26), Milan, Italy, 2026. DOI: 10.1145/3858063.3858904
    - Awarded the QRT Student Travel Award to attend ICAIF '26 (one of 10 recipients).
    - Co-supervising two UvA MSc theses extending this work (multi-agent setting; probabilistic regime detection).

Experience

Research Visit & Collaboration, SIAS Group/Finesse Lab at UvA (Oct 2025 – Ongoing)
University of Amsterdam (UvA) | Amsterdam, Netherlands
- Conducted MSc thesis research with Asst. Prof. Erman Acar in the SIAS group.
- Co-supervising two UvA MSc students who extend the work to a multi-agent setting and to probabilistic regime detection.

Research Assistant – (Scientific Research Project - BAP) (Nov 2024 – Nov 2025)
Bogazici University | Istanbul, Turkiye
- Contributed to the project "Deep Learning-based Exploration of Linguistic Structures and Semantic Entities in Historical Turkish Texts," funded under Bogazici's BAP initiative.
- Assisted in data preparation, feature extraction, and developing NLP models for text processing.

Junior Software Developer (Oct 2023 – May 2024)
Invent Analytics | Istanbul, Turkiye
- Extended and stabilized Invent Mind: hardened the bots, built CI/CD pipelines, and refined prompt engineering to improve performance and reliability. Built a related RFP assistant bot to support the sales team's proposal process.
- Organized and restructured client data using machine learning methods.

Machine Learning Internship (Jun 2023 – Sep 2023)
Invent Analytics | Istanbul, Turkiye
- Built and deployed 'Invent Mind' and 'Invent Echo,' RAG-based conversational systems for internal knowledge sharing, using FAISS for vector similarity search over company documents.
- Developed the retrieval and generation pipeline, integrated Slack, and deployed on AWS Lambda, improving query resolution and developer workflow.

Learning Assistant in CS201 (Introduction to Computer Science) (Mar 2022 – Nov 2022)
Sabanci University | Istanbul, Turkiye

- Conducted weekly recitations and office hours to support student learning of C++ algorithms and problem-solving techniques.

Software Engineering Internship (Jul 2022 – Aug 2022)
GFDS Global Future Design and Solutions Inc. | Ankara, Turkiye

- Developed a conversational AI using NVIDIA's Riva and while doing so, learned the usage of GitHub, Unreal Engine, and Docker while extensively practicing Python and C++.

Academic and Personal Projects

- **MSc Thesis: Internalizing Investment Mandates: Neurosymbolic Reinforcement Learning for Regime-Aware Asset Allocation** (Oct 2025 – Ongoing)
  - Developing a neurosymbolic safe reinforcement learning framework that adapts Probabilistic Logic Policy Gradient (PLPG) to continuous action spaces through a reparameterized Monte Carlo surrogate (PLPG-MC).
  - Combined shielded PPO with HMM-based market regime detection across a 10-ETF universe; reduced constraint violations to ~4.8% versus ~96–98% for an unconstrained baseline.
- **BSc Graduation Project: Intelligent Waste Management System Using Machine Learning and Optimization Algorithms** (Oct 2023 – Jun 2024)
  - Developed a reinforcement learning-based routing algorithm for the Traveling Salesman Problem (TSP) in waste collection.
  - Achieved near-optimal routing results with a 5% deviation from benchmarks, integrating RL with clustering and optimization models.
- **Network Science Research Project & Paper (CS 414)** (Jan 2023 – Jun 2023)
  - Gathered data including a set of Turkish politicians', TV channels', and political party accounts' Twitter followings and used network science metrics to analyze clustering and bias in these account owners.
  - Used Gephi for visualization and metric calculations. Created a research paper and participated in a mini conference inside class.
- **Political Tweet & Bot Detection - Turkish Twitter (CS 412)** (Nov 2022 – Jan 2023)
  - Used XGBoost to accurately detect political tweets and bot accounts; ranked highly on the class model-accuracy leaderboard.

Skills & Interests

- **Languages:** Turkish (Native), English (Full professional)
- **Programming:** Python (Advanced), C++ (Advanced)
- **ML & RL:** PyTorch, PPO (Stable-Baselines3), Hidden Markov Models, XGBoost, Monte Carlo methods, statistical testing
- **Tools & Libraries:** pandas, NumPy, FAISS, AWS (Lambda), Git

Activities

- **President / Vice President, IEEE Computer Society – Sabancı University Branch** (2021 – 2023)
  - Led a club of 60 active members; organized the "Inspire Talks" and "Alumni Talks" series and networking events with alumni and employers.
- **IEEE - Women in Engineering (WIE)** (2021 – 2023)
  - Active member (organization department)
  - Collaborated with other members to organize annual events.
  - Helped organize a big event called "STEMs from Women".